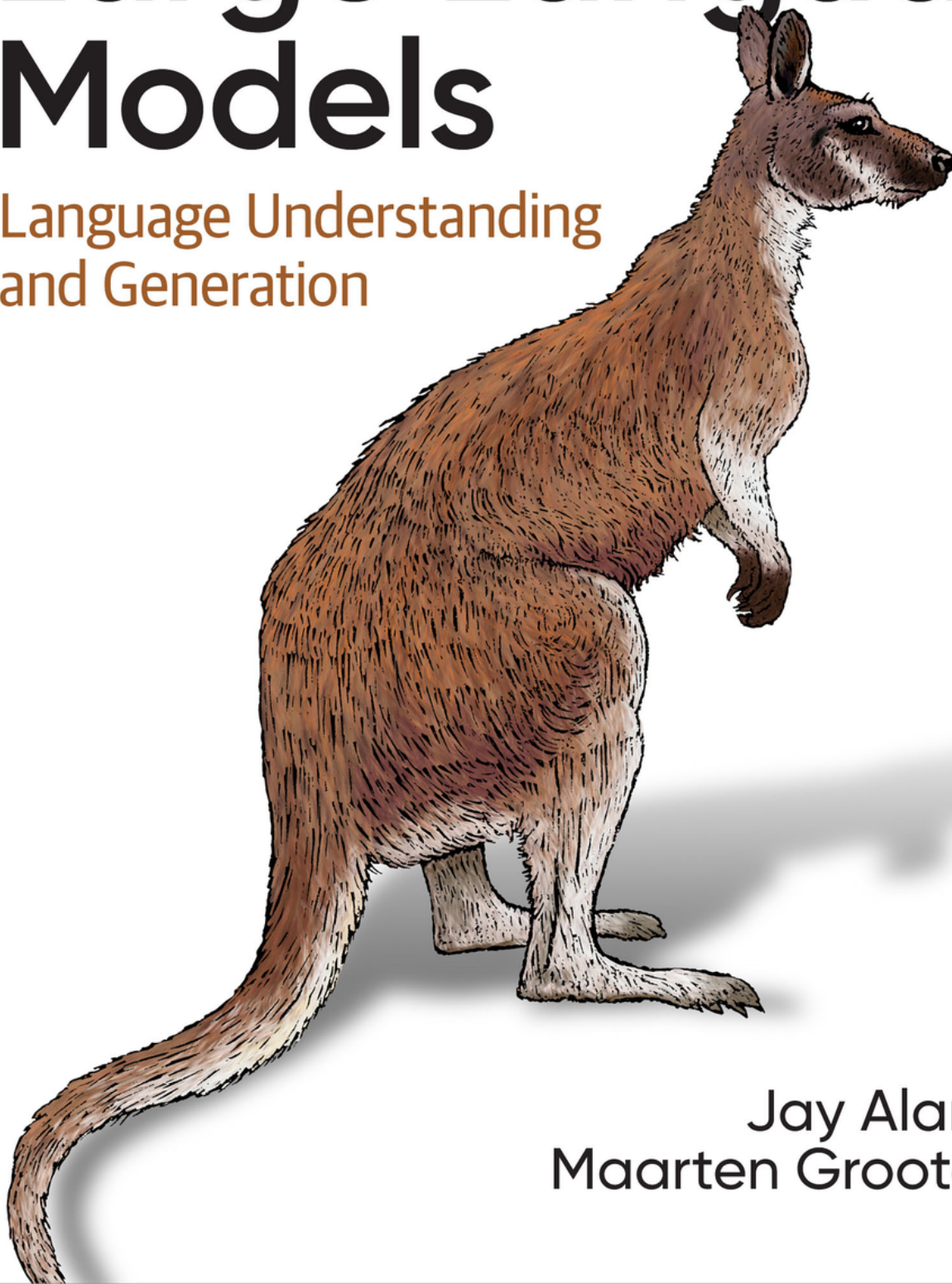


O'REILLY®

Hands-On Large Language Models

Language Understanding
and Generation



Jay Alammar &
Maarten Grootendorst

Praise for *Hands-On Large Language Models*

This is an exceptional guide to the world of language models and their practical applications in industry. Its highly-visual coverage of generative, representational, and retrieval applications of language models empowers readers to quickly understand, use, and refine LLMs. Highly recommended!

—Nils Reimers, Director of Machine Learning at Cohere |
creator of sentence-transformers

Jay and Maarten have continued their tradition of providing beautifully illustrated and insightful descriptions of complex topics in their new book. Bolstered with working code, timelines, and references to key papers, their book is a valuable resource for anyone looking to understand the main techniques behind how Large Language Models are built.

—Andrew Ng, founder of DeepLearning.AI

I can't think of another book that is more important to read right now. On every single page, I learned something that is critical to success in this era of language models.

—Josh Starmer, StatQuest

If you're looking to get up to speed in everything regarding LLMs, look no further! In this wonderful book, Jay and Maarten will take you from zero to expert in the history and latest advances in large language models. With very intuitive explanations, great real-life examples, clear illustrations, and comprehensive code labs, this book lifts the curtain on the complexities of transformer models, tokenizers, semantic search, RAG, and many other cutting-edge technologies. A must read for anyone interested in the latest AI technology!

—Luis Serrano, PhD, Founder and CEO of Serrano
Academy

This book is a must-read for anyone interested in the rapidly-evolving field of generative AI. With a focus on both text and visual embeddings, it's a great blend of algorithmic evolution, theoretical rigor, and practical guidance. Whether you are a student, researcher, or industry professional, this book will equip you with the use cases and solutions needed to level-up your knowledge of generative AI. Well done!

—Chris Fregly, Principal Solution Architect, Generative
AI at AWS

In the heart of the GenAI revolution, this indispensable guide masterfully balances theory and practice, navigating the vast landscape of large language models to equip readers with the knowledge needed for immediate and transformative impact in the field of AI.

—Tarun Narayanan Venkatachalam, AI Researcher,
University of Washington

Timely reading to get hands-on experience with language models.

—Emir Muñoz, Genesys

Hands-On Large Language Models brings clarity and practical examples to cut through the hype of AI. It provides a wealth of great diagrams and visual aids to supplement the clear explanations. The worked examples and code make concrete what other books leave abstract. The book starts with simple introductory beginnings, and steadily builds in scope. By the final chapters, you will be fine-tuning and building your own large language models with confidence.

—Leland McInnes, Researcher at the Tutte Institute for
Mathematics and Computing

Finally, a book that not only avoids superficial coverage of large language models but also thoroughly explores the background in a way that is both accessible and engaging. The authors have masterfully created a definitive guide that will remain essential reading despite the fast-paced advancements in the field.

—Prof. DDr. Roman Egger, CEO of Smartvisions.at and
Modul University Vienna